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Chapter 1

Brownian Motion in deep lubrication

1.1 Introduction

When considering Brownian suspensions, one must remain aware of hydrodynamic interactions at play in the system. Indeed, even at long range, hydrodynamics play a strong role on the behaviour of colloids diffusing in the fluid. Thus, strong care must be taken to ensure that these interactions are accounted for at every range, including the sub-radius range, when a colloid is either close to a confining plane or another colloid. Together, we will see that typically, excellent approximations exist for the so-called far-field and moderate-field regimes, but that the near-field regime requires extra care and involves much more complex physics. We will begin with a full development of the diffusion of a colloid over a hard, infinite boundary, and demonstrate that there exist numerous pitfalls in the formalisation of such a problem. We will then move on to the case of two colloids in a bulk fluid, and present different methods with their respective ranges of validity to do so. Following these two cases, we will follow an experimental study which attempts to formalise the coupling between the wall-colloid and the colloid-colloid effects. Finally, we will discuss how we may go beyond this study, in the deep-lubricated regime, and show how our numerical results compare to the previous study.

1.2 Colloid-plane lubrication theory

First, we wish to understand the influence of boundary below a colloid on the diffusion of this colloid. We take an infinite silica plane, which we will take normal to the vertical z axis, laying at $z = 0$. Over that plane is a fluid of density ρ , viscosity η , and thermal energy $k_B T$ with k_B Boltzmann's constant. As per usual Stokesian assumptions, we will take the fluid as incompressible.

In that fluid, we now submerge a colloidal silica particle of radius a , of the micron scale. Due to the no-slip boundary conditions at contact with the plane and the colloid's

surface, we must modify the effective viscosity of the fluid as the colloid evolves in the z direction. This correction brings not only inhomogeneity along the z -axis but also anisotropy, as the x and y components of the friction coefficient will differ from the z component.

In his seminal work[5], Brenner derives the exact expression for the effective viscosity felt by a sphere moving perpendicularly to a plane wall, $\eta_{\perp}(z)$, as

$$\eta_{\perp}(z) = \eta \frac{4}{3} \sinh(\alpha) \sum_{n=1}^{\infty} \frac{n(n+1)}{(2n-1)(2n+3)} \left[\frac{2 \sinh((2n+1)\alpha) + (2n+1) \sinh(2\alpha)}{4 \sinh^2((n+\frac{1}{2})\alpha) - (2n+1)^2 \sinh^2(\alpha)} - 1 \right], \quad (1.1)$$

with $\alpha = \cosh^{-1}(\frac{z}{a})$. Because Brenner obtains the effective viscosity from full Stokes derivations, it is worth noting that this expression is exact and valid for all values of $z/a > 1$, that is to say up until colloid/plane contact. However, for the sake of efficient numerical computations later on, we will avoid infinite-series computations and prefer the use of Padé approximants [1], which were computed by Honig [11], Bevan and Prieve [3] as

$$\eta_{\perp}(z) \simeq \eta \frac{6z^2 + 2az}{6z^2 - 9az + 2a^2}. \quad (1.2)$$

The parallel effective viscosity, on the other hand, was derived by Faxén [9] and is given by the following expression:

$$\eta_{\parallel}(z) \simeq \frac{\eta}{1 - \frac{9}{16} \frac{a}{z} + \frac{1}{8} \left(\frac{a}{z}\right)^3 - \frac{45}{256} \left(\frac{a}{z}\right)^4 - \frac{1}{16} \left(\frac{a}{z}\right)^5}. \quad (1.3)$$

These expressions suffice for the simulation of a colloid over an infinite plane in far-field to moderate-field regimes.

1.2.1 Summary of potentials in the system

Furthermore, in order to write the correct overdamped Langevin equation for a simulation, we recall that the colloid is also subject to a number of forces, which we will now carefully enumerate as they will be present throughout the remainder parts of this chapter.

Gravitational potential and buoyant mass

$$U_{\text{grav}} = \Delta m g z, \quad (1.4)$$

$$= \frac{4}{3} \pi a^3 (\rho_{\text{colloid}} - \rho_{\text{fluid}}) g z, \quad (1.5)$$

where Δm is the buoyant mass of the colloid, g is the acceleration of gravity, and ρ_{colloid} and ρ_{fluid} are the mass densities of the colloid and the fluid, respectively. We may

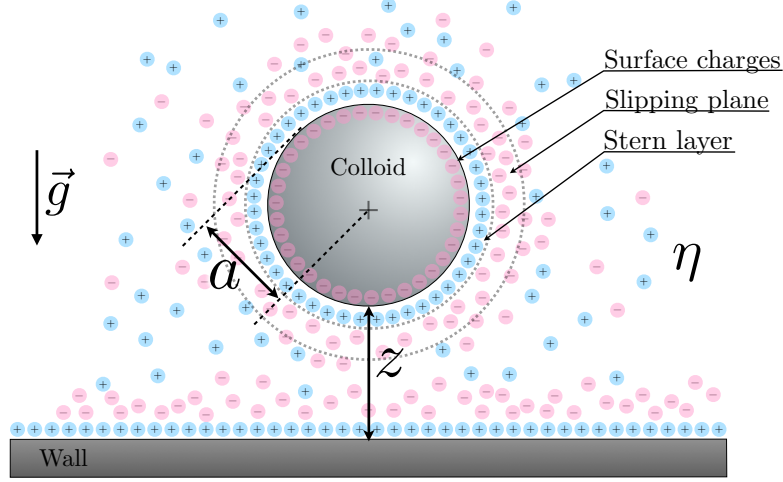


Figure 1.1: Schematic representation of a colloid of radius a at a distance z from an infinite plane. Both surfaces are charged, and thus, a double layer of counter-ions forms around each of them.

rewrite the above Eq. 1.5 as

$$U_{\text{grav}} = \frac{k_B T}{l_B} z, \quad (1.6)$$

where l_B is Boltzmann's length, defined as $l_B = \frac{k_B T}{\Delta m g}$. Boltzmann's length quantifies the distance over which gravitational potential energy is comparable to thermal energy.

Electrostatic potential and double-layer interactions

Because of the amphoteric nature of silica, both the infinite plane and the colloid surfaces are able to ionise when in contact with water (Fig. 1.1). The surfaces themselves become negatively charged, which in turn attract adsorb ions in contact with the surface, in a dense and tightly bound Stern layer. Around this plane, a mobile, ionic "atmosphere" screens the electric potential of the colloid over distance with a characteristic length l_D , for which we provide the derivation below. This diffuse layer, existing for both surfaces at play in the system, is responsible for the electrostatic repulsion between the two bodies (plane and colloid). It is also worth mentioning, for the sake of the following derivation of the electrostatic interaction potential, that within the diffuse layer exists the so-called Slipping plane. This plane separates the fluid that is dragged by the colloid from the fluid that flows around it. Experimentally speaking, this is where one will measure the electrostatic potential of a surface, the ζ potential.

At thermodynamic equilibrium, the local ion concentration in the diffuse layer $c_i(\vec{r})$ of a given species should follow the Boltzmann distribution

$$c_i(\vec{r}) = c_i^\infty \exp \left\{ -\frac{z_i e \psi(\vec{r})}{k_B T} \right\}, \quad (1.7)$$

where $c_{i,\infty}$ is the bulk or infinitely far from surfaces ion concentration, z_i is the valence of the ion, e is the elementary charge, and $\psi(\vec{r})$ is the electrostatic potential at position $\vec{r}$, which satisfies the Poisson equation

$$\nabla^2 \psi(\vec{r}) = -\frac{1}{\varepsilon} \sum_i z_i e c_i(\vec{r}) , \quad (1.8)$$

where ε is the permittivity of the fluid. Substituting the Boltzmann distribution for $c_i(\vec{r})$ into the Poisson equation yields the Poisson-Boltzmann equation, which is a non-linear partial differential equation for $\psi(\vec{r})$.

Thus, combining the above expressions, we get

$$\nabla^2 \psi(\vec{r}) + \frac{e}{\varepsilon} \sum_i z_i c_i^\infty \exp \left\{ \frac{-z_i e \psi(\vec{r})}{k_B T} \right\} = 0 . \quad (1.9)$$

We will only draw our interest to sufficiently dilute solutions in terms of electrolyte concentration $[NaCl]$. In that case, we may Taylor expand the exponential term to obtain

$$\nabla^2 \psi(\vec{r}) + \frac{e}{\varepsilon} \sum_i z_i c_i^\infty \left(1 - \frac{z_i e \psi(\vec{r})}{k_B T} \right) = 0 . \quad (1.10)$$

Since the fluid is electrically neutral

$$\sum_i z_i c_i^\infty = 0 . \quad (1.11)$$

Thus, Eq. 1.10 becomes

$$\nabla^2 \psi(\vec{r}) = \left[\sum_i \frac{z_i^2 e^2 c_i^\infty}{\varepsilon k_B T} \right] \psi(\vec{r}) \quad (1.12)$$

$$= \frac{1}{l_D^2} \psi(\vec{r}) , \quad (1.13)$$

where l_D is Debye's length,

$$l_D = \sqrt{\sum_i \frac{\varepsilon k_B T}{z_i^2 e^2 c_i^\infty}} . \quad (1.14)$$

The above expression can be used to compute the Debye length in a given solution. For the case of a monovalent salt in water, we may keep in mind the simple relation $l_D = 0.304 \sqrt{C}^{-1}$, with C the salt concentration in solution.

Now, we want to compute the electrostatic repulsion potential between a sphere and a plane. To do so, it is convenient to consider the plane as a sphere of infinite radius, thus reducing the problem to a sphere-sphere interaction. The Debye-Huckel expression for

the electrostatic potential around a single charged sphere is given in spherical coordinates by

$$\frac{1}{r^2} \left[\frac{\partial}{\partial r} \left(r^2 \frac{\partial \psi(r)}{\partial r} \right) \right] = \frac{1}{l_D^2} \psi(r) . \quad (1.15)$$

This equation has general solution

$$\psi(r) = K_1 \frac{\exp(\frac{r}{l_D})}{r} + K_2 \frac{\exp(\frac{-r}{l_D})}{r} \quad (1.16)$$

Taking into account that $\lim_{r \rightarrow \infty} \psi(r) = 0$, we identify $K_1 = 0$. Thus, we can write

$$\psi(r) = K_2 \frac{\exp(\frac{-r}{l_D})}{r} . \quad (1.17)$$

We now are left with the identification of K_2 . To do so, we use Gauss's flux theorem, and we finally get

$$\psi(r) = \frac{\sigma a^2}{\varepsilon} \frac{\exp(\frac{a}{l_D})}{1 + \frac{a}{l_D}} \frac{\exp(\frac{-r}{l_D})}{r} , \quad (1.18)$$

where σ is the charge density at the surface of the sphere, a is its radius.

Now that we have the electrostatic potential around a single sphere, we can suppose that the potential between two spheres is simply additive by approximation

$$\psi_{\text{tot}}(r) = \psi_1(r) + \psi_2(r) , \quad (1.19)$$

and with $r = a_1 + a_2 + z$ along sphere line of centers and keeping in mind cylindrical symmetry, the total electrostatic interaction potential along the z-direction becomes

$$U_{\text{tot}}(z) = \frac{4\pi}{\varepsilon} \left(\frac{\sigma_1 a_1^2}{1 + l_D^{-1} a_1} \right) \left(\frac{\sigma_2 a_2^2}{1 + l_D^{-1} a_2} \right) \frac{\exp(-l_D^{-1} z)}{a_1 + a_2 + z} . \quad (1.20)$$

We now have the expression of the interaction potential between two charged spheres. As stated above, we may set one of the two radii to infinity and work with this limit to obtain the interaction potential between a sphere and a plane, as

$$\frac{U_{\text{elec}}}{k_B T} = B e^{-\frac{z}{l_D}} , \quad (1.21)$$

where

$$B = \frac{4\pi}{k_B T \varepsilon} \left(\frac{\sigma a^2}{1 + l_D^{-1} a} \right) l_D \sigma_{\text{wall}} . \quad (1.22)$$

We will furthermore neglect the attractive Van der Waals interactions, which in our cases of study only play a role within a few nanometers, whereas the repulsive interactions take place up to a few tens of nanometers.

Equilibrium distribution

Finally, we may summarize the total potential to which the colloid is subjected when immersed in water above a charged plane as

$$U(z) = U_g + U_{\text{elec}} , \quad (1.23)$$

or, more in detail,

$$\frac{U(z)}{k_B T} = B e^{\frac{-z}{l_D}} + \frac{z}{l_B}; \quad \forall z > 0 , \quad (1.24)$$

as we would not want our colloid to cross the boundary.

As I discussed in the Introduction, the probability to find the colloid at position z given its diffusive behaviour, the gravitational and electrostatic potentials, should follow the Gibbs-Boltzmann equilibrium distribution

$$P_{\text{eq}} = \frac{1}{Z} \exp \left(-\frac{U(z)}{k_B T} \right) . \quad (1.25)$$

This distribution should be normalised, hence the $\frac{1}{Z}$ factor. It is particularly useful to keep the shape of it in mind when designing a simulation, as depending on the effect one wants to probe (very dense suspension, monolayer, or inversely, virtually unbounded colloids in the z -plane), we may play with the parameters to get the desired behaviour.

Fokker-Plank equation with multiplicative noise

Now, in the case of an inhomogeneous and anisotropic diffusion coefficient, the noise in our diffusion equation cannot simply be treated as a Gaussian white noise anymore. Indeed, because its amplitude depends on the diffusivity, and because the diffusivity is neither homogeneous nor isotropic anymore, we will need to include a so-called multiplicative noise in the Langevin equation. This, as we will see, raises the question of the integration of the multiplicative noise, and the intent of the following derivation is to demonstrate the care that must be taken to ensure we obey detailed balance and do not introduce unphysicality by honest mistake.

We begin by taking the following Langevin equation (See the Introduction for its formal presentation), but modify the second term in the right-hand side to include a space-dependent amplitude.

$$dX_t = u(X_t)dt + g(X_t)dB_t , \quad (1.26)$$

where, as a reminder, $u(X_t)$ is the drift from external forces acting upon the colloid, $g(X_t)$ is the space-dependent noise amplitude, and dB_t is a Wiener process, such that $dB_t = \zeta(s)ds$, $\langle dB_t \rangle = 0$ and $dB_t^2 = dt$, which can be seen as the integral of a white noise: it is nowhere differentiable. Integrating Eq. 1.26 over an infinitesimal time step Δt ,

$$X_{t+\Delta t} = X_t + \int_t^{t+\Delta t} u(X_s)ds + \int_t^{t+\Delta t} g(X_s)dB_s . \quad (1.27)$$

It is straightforward to simply integrate the first term of the right-hand side owing to the smoothness of ds using simple methods such as a forward Euler scheme [6]. However, the second term of the right-hand side is known as a stochastic integral, where B_s is a process which has infinite variation. In that case, one cannot simply treat dB_s as a small and deterministic increment, which in turn forbids the use of simple Euler integration. Thus, in order to compute this second integral, we often express it as

$$\int_t^{t+\Delta t} g(X_s)dB_s \equiv g([1-\alpha]X_t + \alpha X_{t+\Delta t}) \Delta W(t) , \quad (1.28)$$

with $\Delta W(t)$ the Wiener increment, as

$$\Delta W(t) = \int_t^{t+\Delta t} dB_s , \quad (1.29)$$

where α encodes where in the time step we want to evaluate the integral. We will need to choose a value for α between 0 (left endpoint) and 1 (right endpoint). Of course, conventions for such integrals exist [12]:

1. The Itô convention: $\alpha = 0$; $\int_t^{t+\Delta t} g(X_s)dB_s \simeq g(X_t)\Delta W$. The system reacts to the noise, there is no feedback within a single timestep. Mathematically, the SDE becomes a *martingale*¹ This means that the process's evolution depends strictly on the previous time step, and not on the history before this time step. However, no matter how convenient and practical this convention may be, it also entails a modification, or rather correction to the chain rule of calculus when changing variables. We discuss this further down, as we will be using this convention for our SDE.
2. The Stratonovich convention: $\alpha = 1/2$; $\int_t^{t+\Delta t} g(X_s)dB_s \simeq g(\frac{X_t+X_{t+\Delta t}}{2})\Delta W$. Stratonovich proposes a way of circumventing the complexities introduced by Itô's convention in the chain rule, breaking the martingale in the process. We now consider a noise which has a finite, non-zero correlation, but this correlation has a characteristic time that is very small compared to the discretisation time step. I will not detail it as much, as I have not really used this convention and am much less aware of its capabilities and pitfalls.

¹I was unaware of the reason for using Itô convention over other counterparts prior to writing the manuscript. However, upon reading literature, I understood how practical working with martingales can be. A martingale has a bounded mean, and, given that we know its history up to time $t' < t$, its expected value at t must be $\mathbb{E}[M_t | \{M_s, s \leq t'\}] = M_{t'}$. In my words, this means that the expected value of a martingale at a given time is the last previously known expected value. The name, however, comes from gambling, a field for which I have exactly zero interest to explore.

Mathematically, it is always possible to identify a Stratonovich process to an Itô process with equivalent solutions. Therefore, it is simply a matter of convenience to choose one over the other. There are, of course, subtleties; and I invite the interested reader to read more on the matter in L. Smith's website [17] from which I liberally adapted some content within this thesis. We'll, for the sake of the development, keep α as a variable for the time being.

We begin by taking Eq. 1.27 to order $\Delta t^{1/2}$

$$X_{t+\Delta t} = X_t + g(X_t)\Delta W(t) \quad (1.30)$$

We can then conveniently develop the $X_{t+\Delta t} - X_t$ term in Eq. 1.28

$$\int_t^{t+\Delta t} g(X_s)dB_s ds = g(X_t + \alpha[X_{t+\Delta t} - X_t]) \Delta W(t) \quad (1.31)$$

$$\approx g(X_t + \alpha g(X_t)\Delta W(t)) \Delta W(t) \quad (1.32)$$

$$\approx g(X_t)\Delta W(t) + \alpha g(X_t)g'(X_t)[\Delta W(t)]^2 \quad (1.33)$$

to order Δt . This in turn gives

$$X_{t+\Delta t} = X_t + [u(X_t) + \alpha g(X_t)g'(X_t)] \Delta t + g(X_t)\Delta W(t) . \quad (1.34)$$

Notice then that the integration of a multiplicative noise brings a correction on the term which depends on Δt , a.k.a. the *drift* term. We want to standardise this expression to easily express the correction to the drift in our equation in order to simulate a trajectory. To do so, we invoke the Fokker-Plank equation yet again

$$\frac{\partial P(X_t, t)}{\partial t} = -\frac{\partial}{\partial X_t} K_1(X_t)P(X_t, t) + \frac{1}{2} \frac{\partial^2}{\partial X_t^2} K_2(X_t)P(X_t, t) , \quad (1.35)$$

where K_1 and K_2 respectively are the first and second moments, which we define as

$$\begin{aligned} K_1(X_t) &= \lim_{\Delta t \rightarrow 0} \frac{\Delta X_t}{\Delta t} \\ &= u(X_t) + \alpha g(X_t)g'(X_t) \end{aligned} \quad (1.36)$$

for the mean, and

$$K_2(X_t) = \lim_{\Delta t \rightarrow 0} \frac{(\Delta X_t)^2}{\Delta t} \quad (1.37)$$

$$= g^2(X_t) \quad (1.38)$$

for the standard deviation. Thus, we can rewrite Eq. 1.35 in those new terms:

$$\frac{\partial P(X_t, t)}{\partial t} = -\frac{\partial}{\partial X_t} \{u(X_t) + \alpha g(X_t)g'(X_t)\} P(X_t, t) + \frac{1}{2} \frac{\partial^2}{\partial X_t^2} \{g(X_t)^2\} P(X_t, t) . \quad (1.39)$$

Derivation of the stochastic drift term

We would now like to apply this knowledge to compute the new stochastic drift term one must add in the dt order to achieve physicality in our system. We begin again with the Langevin equation of a colloid sedimenting above a plane

$$dz = \frac{F(z)}{\gamma_{\perp}(z)}dt + \sqrt{2D_{\perp}(z)}dB_t . \quad (1.40)$$

In that case, taking Eq. 1.36, we express the mean velocity as

$$\left\langle \frac{\Delta z}{\Delta t} \right\rangle \equiv \bar{v}_d = \frac{F(z)}{\gamma_{\perp}(z)} + \alpha \frac{dD_{\perp}(z)}{dz} . \quad (1.41)$$

where $F(z)/\gamma_{\perp}(z)$ is our term of interest, and it is worth noting it contains the complexities of the following derivations and is, in no way, trivial and simply deriving from potentials.

We begin by taking the steady-state of Eq. 1.39 and differentiating with respect to time. By definition,

$$\frac{d}{dz} \left\{ -u(z) - \alpha g(z)g'(z) + \frac{1}{2} \frac{d}{dz} g(z)^2 \right\} P_{\text{steady-state}}(z) = 0 . \quad (1.42)$$

with $u(z) = \frac{F}{\gamma_{\perp}}$ and, because $\lim_{z \rightarrow \infty} P_{\text{steady state}}(z) = 0$ and $\lim_{z \rightarrow \infty} \frac{dP_{\text{steady state}}(z)}{dz} = 0$,

$$(-u(z) - \alpha g(z)g'(z))P_{\text{steady-state}}(z) + \frac{1}{2} \frac{d}{dz} \{g(z)^2 P_{\text{steady-state}}(z)\} = 0 . \quad (1.43)$$

We may now substitute $A(z) = g^2(z)P_{\text{steady-state}}(z)$, which yields

$$\frac{dA(z)}{dz} - 2 \frac{u(z) + \alpha g(z)g'(z)}{g^2(z)} A(z) = 0 , \quad (1.44)$$

which integrates as

$$\ln(A(z)) = \ln(g^2(z)) + \ln(P_{\text{steady state}}) = \int \frac{u(z') + \alpha g'(z')g(z')}{g^2(z')/2} dz' . \quad (1.45)$$

And thus,

$$P_{\text{steady-state}}(z) = \frac{1}{g^2(z)} \exp \left\{ 2 \int_0^z \frac{u(z') + \alpha g'(z')g(z')}{g^2(z')} dz' \right\} , \quad (1.46)$$

We note that we may express $\ln(g^2(z)) = \int \frac{2g'(z')g(z')}{g^2(z')} dz'$, and

$$U_{\text{steady-state}}(z) = -k_B T \int \frac{u(z') + (\alpha - 1)g(z')g'(z')}{g^2(z')/2} dz' . \quad (1.47)$$

Remembering that $F_{\text{steady state}} = -\frac{dU_{\text{steady state}}}{dz}$, we get

$$\frac{F_{\text{steady state}}}{\gamma_{\perp}(z)} = \frac{F(z)}{\gamma_{\perp}(z)} + (\alpha - 1)g(z)g'(z) . \quad (1.48)$$

From the Lanvegin equation Eq. 1.40, we note that $g(z) = \sqrt{2D_{\perp}(z)}$, and $F(z) = \frac{u(z)}{\gamma_{\perp}}$, and we also keep in mind that $D_{\perp} = \frac{k_{\text{B}}T}{\gamma_{\perp}}$. With the Itô convention $\alpha = 0$, we may now write the total drift of Eq. 1.51 as

$$\bar{v}_{\text{d}} = \frac{1}{\gamma_{\perp}(z)} \frac{dU_{\text{steady-state}}(z)}{dz} + \frac{dD_{\perp}(z)}{dz} . \quad (1.49)$$

By identification, we then recover the deterministic drift velocity emerging from external forces

$$v_{\text{d}} = \frac{k_{\text{B}}T}{\gamma_{\perp}(z)} \left[-\frac{1}{l_{\text{D}}} B \exp\left(-\frac{z}{l_{\text{D}}}\right) + \frac{1}{l_{\text{B}}} \right] , \quad (1.50)$$

whereas the stochastic drift, in the Padé approximation, reads

$$v_{\text{stochastic drift}} = 2D_0 a \frac{2a^2 + 12az + 21z^2}{(2a^2 + 9az + z^2)^2} . \quad (1.51)$$

In the latter parts of this chapter, I will use an algorithm that is specifically made to handle Brownian motion in confined geometries. It is worth keeping in mind the present derivation holds for this algorithm, which is why I deemed it relevant to be included in this thesis.

Simple Langevin simulation of a confined colloid

A very usual task in my group, thanks to the theses produced and defended by Maxime Lavaud [14] and Elodie Millan [15], is to simulate the system I just spent a few pages describing throughly. Thought not strictly part of my doctoral work (it was more of a Master's project under Maxime and Elodie's supervisions), I briefly include it here both to pay my respects to them and to show what can be done with minimal computations and basic code. I invite the interested reader to check their theses for ready-to-run, Cython²-compiled codes [7], and I provide minimal versions here (**TODO : make a little jupyterlab for the reader to play around online. Think about whether to make one file per fig or a whole working example with simu and amalysis in the same file like Ryker did for libmobility in zaragoza.**).

$$x_i = x_{i-1} + \sqrt{2D_{\parallel}(z_{i-1})\Delta t} w_{x_i} , \quad (1.52)$$

$$z_i = z_{i-1} + \bar{v}_{\text{d}}(z_{i-1})\Delta t + \sqrt{2D_{\perp}(z_{i-1})\Delta t} w_{z_i} , \quad (1.53)$$

²Cython refers both to a compiler and a programming language that is based on Pyrex. Owing to its high level of abstraction, Cython is extremely easy to write, maintain and comprehend, all while reaching the optimization capabilities of pure C, C++ or Fortran code. When I developed algorithms myself from scratch, I mostly favoured Python, and implemented computationally heavier loops in Cython.

where w_{x_i} and w_{z_i} are two independent Gaussian white noises of zero mean and Δt variance, and where $\bar{v}_d(z)$ is the average drift velocity of the colloid at position z as computed in Eq. 1.51, which includes both deterministic and stochastic contributions as per equation 1.41.

We then carefully choose a time step such that ... **Reference the introductory part. Write this part when I have rewritten the code. to do : -explanation of code, figure of traj, peq, MSD.**

1.2.2 Colloid-colloid lubrication theory

Colloids displace fluid when they diffuse. Intuitively, it is very easy to understand that, if you were to place two colloids or more in an otherwise unbounded fluid, their dynamics should couple in *some way*. Naturally, this question has been investigated through and throughout already, and I will start by presenting the problem in terms of mobility $\mathbf{M}$ instead of diffusion. I will explain how Oseen formalism [16] can be used to probe at this coupling. I'll, however, mitigate the accuracy of the method, and present a second way of computing more accurately (and more expensively) the same quantities.

The mobility problem

We begin with N colloids submerged in an unbounded bulk fluid. We work at low Reynolds number ($\text{Re} \ll 1$) and neglect inertial effects. In that case, the Stokes equation become linear

$$-\nabla p(\mathbf{r}) + \eta \nabla^2 \mathbf{u}(\mathbf{r}) = -\mathbf{F} \delta(\mathbf{r}) , \quad (1.54)$$

$$\nabla \cdot \mathbf{u} = 0 \quad (1.55)$$

where ∇p is the pressure gradient in the fluid, that is to say, the force density due to pressure differences in the fluid; $\mu \nabla^2 \vec{u}$ is the viscous term that controls how the fluid diffuses momentum through itself with μ the fluid viscosity and $\vec{u}$ the fluid velocity field; $F\delta(\vec{r})$ is the point force at origin with δ the Kronecker function; and the second equation denotes fluid incompressibility.

The velocity vector can be written as a product of the mobility tensor $\mathbf{M}$ and the force vector $\mathbf{f}$

$$\mathbf{v} = \mathbf{M} \cdot \mathbf{f} , \quad (1.56)$$

where the mobility matrix $\mathbf{M}$ is of dimension $6N \times 6N$ and has the structure

$$\mathbf{M} = \begin{bmatrix} \mu_{11}^{tt} & \cdots & \mu_{1N}^{tt} & \mu_{11}^{tr} & \cdots & \mu_{1N}^{tr} \\ \vdots & \ddots & \vdots & \vdots & \ddots & \vdots \\ \mu_{N1}^{tt} & \cdots & \mu_{NN}^{tt} & \mu_{N1}^{tr} & \cdots & \mu_{NN}^{tr} \\ \mu_{11}^{rt} & \cdots & \mu_{1N}^{rt} & \mu_{11}^{rr} & \cdots & \mu_{1N}^{rr} \\ \vdots & \ddots & \vdots & \vdots & \ddots & \vdots \\ \mu_{N1}^{rt} & \cdots & \mu_{NN}^{rt} & \mu_{N1}^{rr} & \cdots & \mu_{NN}^{rr} \end{bmatrix} \quad (1.57)$$

and $\mathbf{f} = \mathbf{F}_1, \dots, \mathbf{F}_N, \mathbf{T}_1, \dots, \mathbf{T}_N$ with $\mathbf{F}$ the forces, and $\mathbf{T}$ the torques. Subsequently,

$$\mathbf{v}_i = \sum_j (\mu_{ij}^{tt} \cdot \mathbf{F}_j + \mu_{ij}^{tr} \cdot \mathbf{T}_j) \quad , \quad (1.58)$$

$$\mathbf{w}_i = \sum_j (\mu_{ij}^{rt} \cdot \mathbf{F}_j + \mu_{ij}^{rr} \cdot \mathbf{T}_j) \quad , \quad (1.59)$$

with $\mathbf{v}_i$ the translational and $\mathbf{w}_i$ rotational velocities of the colloid indexed i . It is worth mentioning that in our problems, the mobility matrix must be symmetric positive definite [10], which greatly simplifies its structure.

The Oseen tensor

We consider the case of a colloid moving in an infinite and quiescent fluid. By effectively zooming out, we may approximate this colloid as a point force $\mathbf{F}_1$ acting upon the fluid at a given position $\mathbf{r}_1$.

In that case, the velocity field $\mathbf{u}(\mathbf{r})$ is written

$$\mathbf{u}(\mathbf{r}) = \mathbf{G}(\mathbf{r} - \mathbf{r}_1) \cdot \mathbf{F} \quad (1.60)$$

where $\mathbf{G}$ is the Oseen tensor. Because $\mathbf{u}(\mathbf{r})$, is the fundamental solution of the Stokes equation (for a point force, that is), it is referred to as a "Stokeslet" [13], and the Oseen tensor $\mathbf{G}((\mathbf{r}) - \mathbf{r}_1)$ is the Green's function to the Stokes equations. We will, in all of the following derivations, not take into account rotation of the colloids, that we will consider as perfect spheres. We will, however, keep track of their orientation anyway, for the sake of the derivation and later on to check numerical accuracy. With this, we may now express the flow field generated for a continuous force in space as the superposition of Green's functions

$$\mathbf{u}(\mathbf{r}) = \int \mathbf{G}(\mathbf{r} - \mathbf{r}') \cdot \mathbf{f}(\mathbf{r}') d^3 r' \quad . \quad (1.61)$$

Solving the Stokes equation, and taking into account symmetries ($\frac{1}{4\pi}$ factor) and incompressibility ($\frac{1}{2}$ factor), we then obtain

$$\mathbf{G}(\mathbf{r}) = \frac{1}{8\pi\eta r} (\mathbf{I} + \hat{\mathbf{r}}\hat{\mathbf{r}}) \quad . \quad (1.62)$$

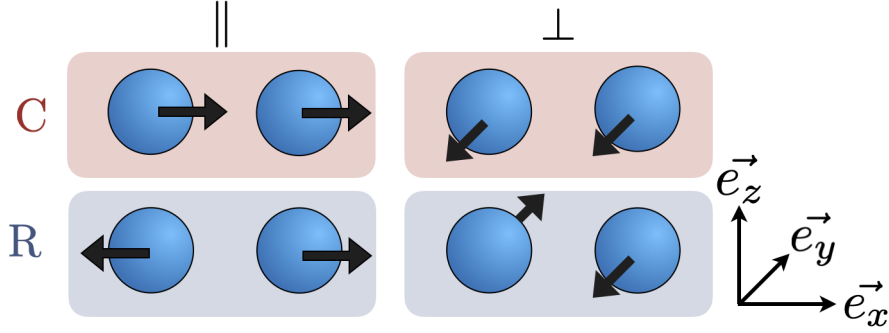


Figure 1.2: Diagram of the different modes of motion of interest for two Brownian colloids. For the sake of simplicity and continuity with the rest of the chapter, we do not consider the vertical direction.

Now, if we want to solve for two colloids at positions $\mathbf{x}$ and $\mathbf{r}_j$ in interaction in the fluid, we simply obtain

$$\mathbf{G}(\mathbf{x} - \mathbf{r}_j) = \frac{1}{8\pi\eta} \left(\frac{1}{|\mathbf{x} - \mathbf{r}_j|} + \frac{(\mathbf{x} - \mathbf{r}_j)(\mathbf{x} - \mathbf{r}_j)}{|\mathbf{x} - \mathbf{r}_j|^3} \right) \quad (1.63)$$

A diffusion tensor between two colloid is expressed in its full form as

$$\langle \Delta r_{i,\alpha}(\Delta t) \Delta r_{j,\beta}(\Delta t) \rangle = 2D_{i\alpha,j\beta} \Delta t, \quad (1.64)$$

where the α and β components correspond respectively a sphere moving along the spheres lines of centers ($\parallel$) or perpendicularly to their lines of center ($\perp$) (Fig. 1.2.2) [8].

$$D_{i\alpha,j\beta} = k_B T G_{\alpha\beta}(\vec{r}_1 - \vec{r}_2). \quad (1.65)$$

Thus, the diagonal terms of the tensor are simply the self-diffusion of a colloid, whereas the off-diagonal blocks are the actual cross-diffusion we are interested in. By diagonalising the tensor, we retrieve

$$\frac{D_{\perp}^{C,R}}{2D_0} = 1 \pm \frac{3}{4} \frac{a}{r} + \mathcal{O}\left(\frac{a^3}{r^3}\right), \quad (1.66)$$

$$\frac{D_{\parallel}^{C,R}}{2D_0} = 1 \pm \frac{3}{2} \frac{a}{r} + \mathcal{O}\left(\frac{a^3}{r^3}\right), \quad (1.67)$$

where positive and negative corrections correspond respectively to the collective and relative modes of motion. As one may notice, hydrodynamic interactions are very long-range with respect to the colloid size and decay in $1/r$.

Now, as I discuss in this subsection, the Oseen approximation does not work for near-field interactions. Referring specifically to Eq. 1.67, our expressions are only valid

down to $r \geq 10a$. Beyond that limit, the approximation becomes first incorrect, and then unphysical, with negative diffusion coefficients arising.

Luckily, because of its construction, the Oseen approximation is actually the first order of a more robust correction, which I will detail in the next subsection.

The Rotne-Prager-Yamakawa tensor

To remedy the issues of the Oseen approximations mentioned above, we introduce the Rotne-Prager-Yamakawa (RPY) tensor, which incorporates finite size particles into the hydrodynamic interactions, at the pairwise level. This correction is, therefore, not a full N-body solution (which is anyway not solvable in a closed, analytic form, and would be a numerical nightmare to compute efficiently). I would put it, simply, as a finite-size correction to the Oseen approximation. Explicitly, for the two-colloid case

$$\mu_{12}^{\text{RPY}}(\mathbf{r}) = \frac{1}{8\pi\eta\mathbf{r}} \left[\left(1 + \frac{2a^2}{3r^2}\right) \mathbf{I} + \left(1 - \frac{2a^2}{r^2} \frac{\mathbf{r}\mathbf{r}}{r^2}\right) \right]. \quad (1.68)$$

This expression also ensures positive-definiteness of the mobility tensor. We will see, later down, that it is used in a modified form to account for the existence of boundaries in the fluid.

Should I detail this more guys ? I feel like, compared to the rest of my derivations, it is a little bit hollow, but on the other side I have not exactly derived it ever, nor have I felt the need to.

1.2.3 Coupling colloid-wall and colloid-colloid lubrication theories

Naturally, the next step is to couple both lubrication effects together. Luckily, such study has already been carried out both theoretically and experimentally by Dufresne et al. [8]. In this article, the authors discuss the modification on colloid cross-diffusion due to wall and colloid-colloid-induced lubrication effects. The following setup, depicted in Figure 1.2.3, is used: two colloids of radius $a = 0.485 \pm 0.025\mu\text{m}$ (Duke Scientific lot 21024) are submerged in a water-salt solution of 2mM [NaCl] and viscosity $\eta = 0.817\text{cP}$. The water-salt layer is $140 \pm 2\mu\text{m}$ thick, and sealed between a microscope slide and a No. 1 coverslip, with the system fully sealed with UV-cured epoxy to prevent evaporation of the fluid and fluid flows in the bulk. The system is maintained at $T = 29.00 \pm 0.01^\circ\text{C}$. In these conditions, the colloid self-diffusion is found to be $D_0 = \frac{k_B T}{6\pi\eta a} = 0.550 \pm 0.028\mu\text{m}^2/\text{s}$.

The authors then trap the two colloids with a single optical tweezer, by alternating the position of the focal plane at the two positions of interest at a frequency of 200Hz using a galvanometer-driven mirror. Then, by blocking the beam for $t = 83\text{ms}$, they retrieve the sphere positions over five video frames only to ensure minimal out-of-focal displacement. From there on, authors carry out experiments over a grid of starting positions in (r, h) space, with r the center-to-center distance between colloids and $2\mu\text{m} < r < 10\mu\text{m}$, and h the colloid center-to-wall distance, with $1\mu\text{m} < h < 30\mu\text{m}$. By reactivating the trapping, they can repeat the experiment with perfect reproducibility for each starting position of interest in their system.

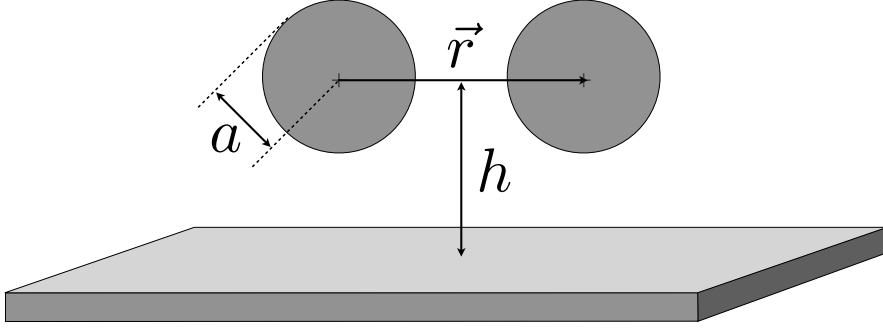


Figure 1.3: Experimental setup of Dufresne et al. [8] for the study of two colloids diffusing over a plane wall.

As discussed in the previous sections, colloids in a same bath should see their motion coupled through hydrodynamic interactions, but the coupling to the wall itself should also modify their expected diffusion. To investigate these coupling effects, the authors decompose the trajectories of the colloids into cooperative and relative motions, respectively $\vec{\rho} = \vec{r}_1 + \vec{r}_2$ and $\vec{r} = \vec{r}_1 - \vec{r}_2$ along ($\parallel$) and perpendicular ($\perp$) directions to the line of centers of the colloids (Fig. ??). From there, they fill displacement histograms for each mode of motion at each of the five time steps taken into consideration and retrieve the effective diffusion coefficients from the standard deviation of each histogram, as

$$\langle \Delta r_{i,\alpha}(\tau) \Delta r_{j,\beta} \rangle = 2D_{i\alpha,j\beta}\tau, \quad (1.69)$$

with $i, j \in \{1, 2\}$ and $\alpha, \beta \in \{\parallel, \perp\}$, where τ is the elapsed time, and where $\langle . \rangle$ denotes an ensemble average.

A typical result of the experiment for a given set of (r, h) is shown in Figure 1.2.3. There, MSDs are extracted from displacement histograms at each of the five time lags into consideration. From there, authors can extract the effective cross-diffusion coefficient.

For comparisons purposes with their experimentally retrieved cross-diffusions in the four possible modes, the authors then compute by different means the theoretical expected value of the cross-diffusion coefficients.

Now, in the presence of a wall, one must take extra care to ensure no-slip boundary conditions at the wall as well as no fluid penetration through the wall itself. To do this, one may use the single-colloid method of images [4], where a Stokeslet (S), as Source Doublet (D) and a Stokeslet Doublet (SD) of opposite strength to the Stokeslet of interest at $\vec{r}_j$ above the wall is placed at equal distance below the wall, at $\vec{R}_j = \vec{r}_j - 2h\vec{e}_z$. The goal of the mirror Stokeslet is to cancel most of the normal penetration at the wall. The Source Doublet ensures no-slip at the wall, and the Stokeslet Doublet cancels remaining penetration. Then, the flow at the position of the image colloid is written

$$G_{\alpha\beta}^W(\vec{x} - \vec{R}_j) = -G_{\alpha\beta}^S(\vec{x} - \vec{R}_j) + 2h^2 G_{\alpha\beta}^D(\vec{x} - \vec{R}_j) - 2h G_{\alpha\beta}^{SD}(\vec{x} - \vec{R}_j), \quad (1.70)$$

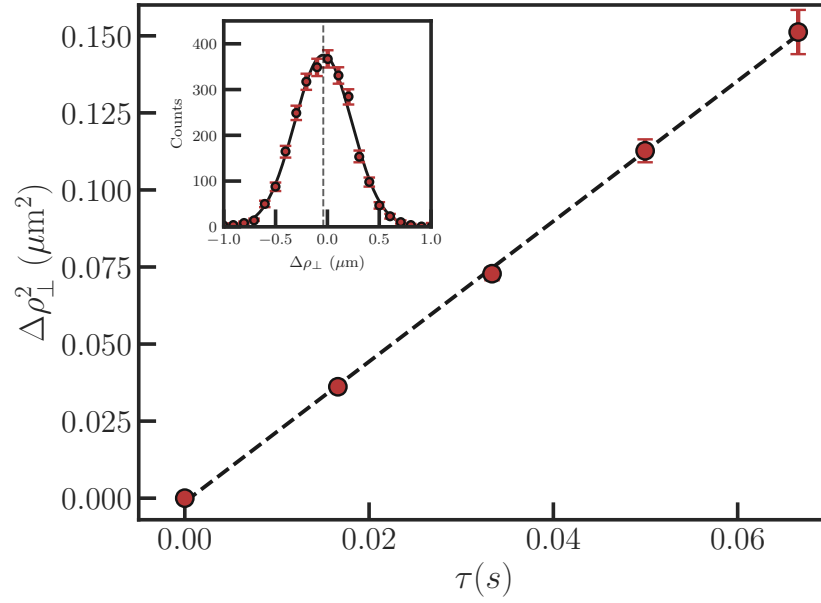


Figure 1.4: Adapted from [8]. Measured pair diffusion for the relative perpendicular mode of motion of the two colloids. The upper inset shows the histogram of all binned displacements for a given time step, for $h = 25.5 \pm 0.7 \mu\text{m}$ and $r = 7.00 \pm 0.25 \mu\text{m}$. This process is repeated over all time steps and each histogram is fitted with a Gaussian curve to recover the root mean square displacement $\Delta\rho(r, h, \tau)$ to recover the MSD of the main plot. The effective cross-diffusion coefficient for the perpendicular mode is then extracted from the slope of the MSD.

with

$$G_{\alpha\beta}^D(\vec{x} - \vec{R}_j) = \frac{1}{8\pi\eta} (1 - 2\delta_{\beta z}) \frac{\partial}{\partial R_{j,\beta}} \left(\frac{(\vec{x} - \vec{R}_j)_\alpha}{|\vec{x} - \vec{R}_j|^3} \right), \quad (1.71)$$

$$G_{\alpha\beta}^{SD}(\vec{x} - \vec{R}_j) = (1 - 2\delta_{\beta z}) \frac{\partial}{\partial R_{j,\beta}} G_{\alpha z}^S(\vec{x} - \vec{R}_j). \quad (1.72)$$

which yields the two following modes of motion for the cross-diffusion coefficients

$$\frac{D_z}{D_0} = 1 - \frac{9}{8} \frac{a}{r} + \mathcal{O}\left(\frac{a^3}{r^3}\right), \quad (1.73)$$

$$\frac{D_{xy}}{D_0} = 1 - \frac{9}{16} \frac{a}{r} + \mathcal{O}\left(\frac{a^3}{r^3}\right). \quad (1.74)$$

One may notice the resemblance between the correction terms in the above expressions and the corrections in the Faxén formula from equation 1.3, which is not surprising as both corrections are due to the same wall-induced hydrodynamic effects. However, these expressions are only valid in the far-field regime, and fail to capture the lubrication effects in the near-field regime, which is precisely the regime of interest for our simulations.

The authors then propose an intermediate regime, first by naively superimposing the Oseen and Blake approximations as $D_\psi^{-1}(r, h) = D_\psi^{-1}(r) + [D_{xy}^{-1}(h) - D_0^{-1}] / 2$ (in dash lines in Figure 1.2.3), which is a crude approximation which fails to capture the behaviour of the cross-diffusion for colloids below 50 radii from the plane. The authors then propose that the colloid must interact with its image, its neighbour, and its neighbour's image. In that case, the cross-diffusion is given by

$$\frac{D_\perp^{C,R}}{2D_0} = 1 - \frac{9}{16} \frac{a}{r} \pm \frac{3}{4} \frac{a}{r} \left(1 - \frac{1 + \frac{3}{2}\xi}{(1 + \xi)^{3/2}} \right) \quad (1.75)$$

$$\frac{D_\parallel^{C,R}}{2D_0} = 1 - \frac{9}{16} \frac{a}{r} \pm \frac{3}{2} \frac{a}{r} \left(1 - \frac{1 + \xi + \frac{3}{2}\xi^2}{(1 + \xi)^{5/2}} \right) \quad (1.76)$$

with $\xi = 4h^2/r^2$ and up to $\mathcal{O}\left(\frac{a^3}{r^3}\right)$ and $\mathcal{O}\left(\frac{a^3}{h^3}\right)$ (in solid lines in Figure 1.2.3).

Furthermore, as noted by the authors, in the very far-field regime, the Oseen approximations should be recovered numerically and experimentally, whereas they expect the above expressions to hold in the very near-field regime as the wall hinders coupling between colloid motions.

In the following parts, we will go about confirming the above predictions numerically, but we will push the study much deeper into the lubrication regimes to investigate its validity once we cannot ignore finite-size effects anymore in the lubrication approximations.

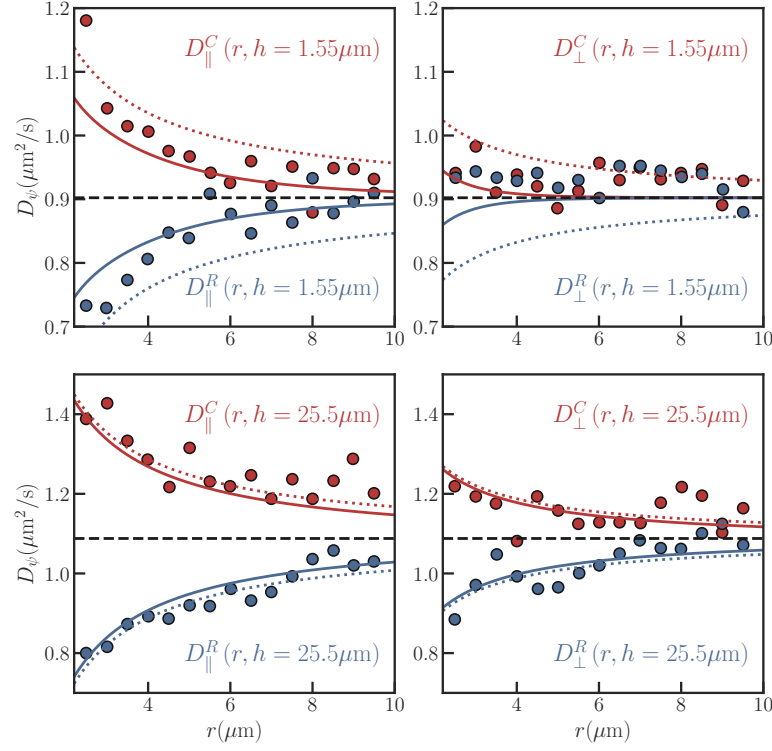


Figure 1.5: Adapted from [8]. Measured cross-diffusion coefficients for the relative perpendicular and parallel modes of motion of the two colloids, as a function of the center-to-center distance r between the colloids, for different values of the center-to-wall distance h . The dash lines correspond to the naive superposition of Oseen and Blake approximations, while the solid lines correspond to the more accurate approximation taking into account the interaction with the image colloid.

1.3 Simulating dense suspensions in close confinement

Naturally, researchers have been pondering on how to simulate dense Brownian suspensions for a few decades already [2]. However, the numerical complexity of such a method (and namely, how to efficiently solve for a multiplicative noise and, a fortiori, how to compute efficiently the square root of a diffusion or mobility tensor without computational cost blowups)...

Todo : finish this when i have had time to prepare a code, it will be clearer in my mind. add results and comparisons.

The lubrication-corrected version of the RigidMultiBlobsWall (RBW) algorithm [18] (here on GitHub, as `stochastichydrotools/rigidmultiblobswall`) aims at filling the gap ³ that is forbidden by the RPY approximation. As I mentioned earlier, it is computationally expensive to fully resolve the fluid and every boundary precisely. Therefore, the RBW algorithm proposes :

- In the far-field, to use the RPY approximation anyway, because it is exact in that regime;
- In the near-field, to replace the RPY approximation by subtraction of its contribution and addition of asymptotical corrections to the mobility matrix.

In an effort of convenience, because the resistance matrix is additive, we will introduce it as $\mathbf{R} = \mathbf{M}^{-1}$ the inverse of the mobility matrix.

We take again a suspension of N colloids, and enforce a no-slip boundary below on the $z = 0$ plane. Formally, we write the new, corrected mobility matrix of the whole system as

$$\tilde{\mathbf{M}} = [\mathbf{R} + \mathbf{R}_{\text{lub}}^{\text{sup}} - \mathbf{R}_{\text{RPY}}^{\text{sup}}]^{-1} , \quad (1.77)$$

where $\mathbf{R}$ is constructed from pairwise RPY approximations, $\mathbf{R}_{\text{lub}}^{\text{sup}}$ is the new, asymptotical corrections that are added to the resistance of nearby colloids, and $\mathbf{R}_{\text{RPY}}^{\text{sup}}$ is the erroneous RPY corrections that $\mathbf{R}$ includes due to the RPY not correcting the close-field appropriately. *Todo : when I work again on the code, go over this, as right now i don't really remember it...*

³no pun intended

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